MOBILITY MANAGEMENT

(Mobile Computing)

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1. Definition

Mobility Management is a technique used in **mobile computing and cellular networks** to **track and manage the location of mobile devices** (phones, laptops, tablets) while they move from one network area (cell) to another.

It ensures that a **mobile user can continue communication** (calls, internet, messages) even when moving between different locations.

Simple Definition (Exam Oriented):

Mobility Management is the process of **tracking the location of mobile devices and maintaining communication when they move between different network cells.**

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2. Main Functions of Mobility Management

Mobility Management mainly performs **two important functions**:

Function	Explanation
Location Management	Keeps track of the current location of the mobile device in the network.
Handoff Management	Transfers an ongoing call or data session from one base station to another when the user moves.

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3. Location Management

Location Management helps the network **find the mobile user** in the system.

It consists of two main processes:

3.1 Location Update

When a user moves to a **new location area**, the mobile device sends an **update message to the network** so that the system knows the new location.

3.2 Paging

When someone calls the user, the network **searches for the mobile device by sending paging signals** to different cells.

Example:

If you travel from **Mumbai to Pune**, the network updates your location so that calls and messages can reach you.

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4. Handoff Management (Handover)

Handoff occurs when a **mobile user moves from one cell tower to another during an active call or data session**.

The network automatically transfers the connection from the **old base station to the new base station** without interrupting the call.

Example:

If you are **talking on the phone while traveling in a car**, your call switches between towers automatically without disconnecting.

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5. Types of Handoff

Type	Explanation
Hard Handoff	The old connection is terminated before establishing the new connection.
Soft Handoff	The mobile device connects to multiple base stations simultaneously for a short time.

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6. Components Used in Mobility Management

Component	Description
Mobile Station (MS)	User device such as mobile phone or laptop.

Base Station (BS)	Communicates with mobile devices and provides network connectivity.
Home Location Register (HLR)	Stores permanent subscriber information.
Visitor Location Register (VLR)	Stores temporary information about users currently in the area.

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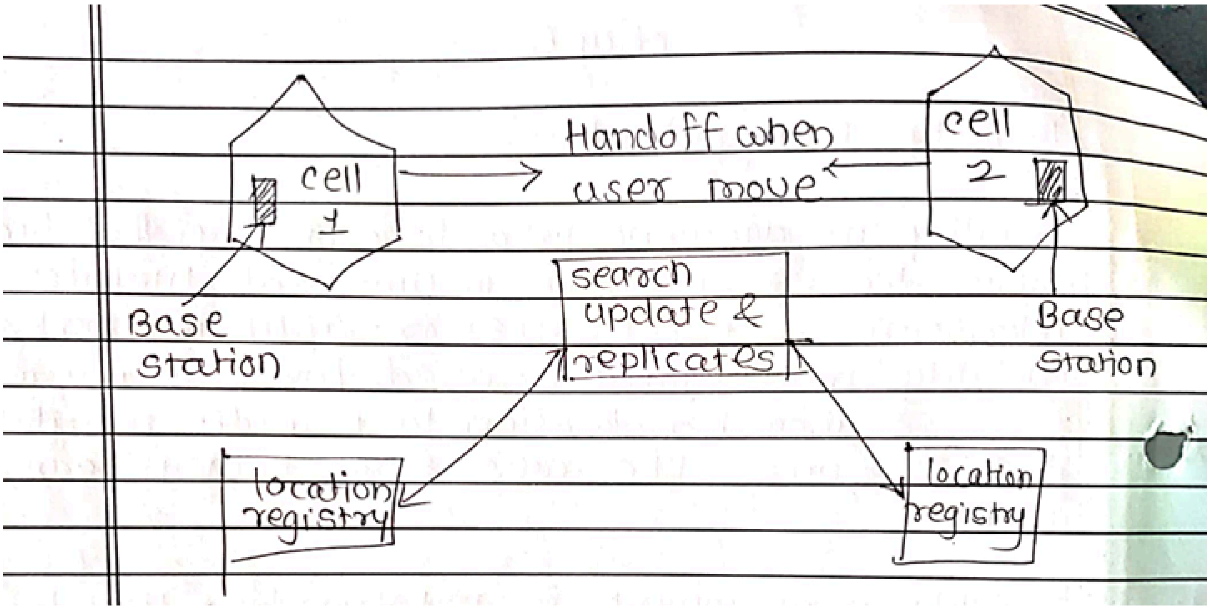
7. Advantages of Mobility Management

- Provides **continuous communication while moving**
- Reduces **call drops**
- Enables **efficient network management**
- Improves **overall user experience**



IP MOBILITY

(Mobile Computing)

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1. Definition

IP Mobility refers to the ability of a **mobile device to move from one network to another while maintaining the same IP address and continuing ongoing internet communication.**

Normally, in the Internet, an **IP address is linked to a specific network location**. When a device moves to a different network, its IP address changes, which may interrupt ongoing connections.

However, with **IP Mobility**, a device can **change networks without changing its permanent IP address**, ensuring that communication continues without interruption.

Exam Definition:

IP Mobility is a technique that allows a **mobile node to maintain the same IP address while moving across different networks and continue communication without disruption.**

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2. Need for IP Mobility

Without IP Mobility:

- When a device moves to another network, its **IP address changes**
- Existing **connections break**
- Communication must restart

With IP Mobility:

- Device keeps the **same IP address**
- Ongoing **communication continues smoothly**

Example:

If you start downloading a file using **Wi-Fi** and then switch to **mobile data**, IP mobility ensures that the download continues without interruption.

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3. Components of Mobile IP

Component	Description
Mobile Node (MN)	The mobile device that moves between different networks

Home Agent (HA)	A router in the home network that keeps track of the mobile node's location
Foreign Agent (FA)	A router in the visited network that assists the mobile node
Care-of Address (CoA)	A temporary IP address assigned to the mobile node in the foreign network

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4. Working of IP Mobility (Mobile IP)

Step 1 – Mobile Node Movement

The **mobile node moves from its Home Network to a Foreign Network**.

Step 2 – Care-of Address Allocation

The foreign network provides the mobile node with a **Care-of Address (temporary IP address)**.

Step 3 – Registration

The mobile node **registers its Care-of Address with the Home Agent** so the network knows its current location.

Step 4 – Packet Forwarding

The **Home Agent receives packets intended for the mobile node** and forwards them to the **Care-of Address** in the foreign network.

This packet forwarding process is known as **tunneling**.

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5. Important Terms

Term	Meaning
Home Network	The original network where the mobile node is registered
Foreign Network	The network where the mobile node currently moves
Tunneling	The process of forwarding packets from the Home Agent to the Care-of Address
Care-of Address (CoA)	Temporary address assigned to the mobile node in the foreign network

6. Advantages of IP Mobility

- Provides **continuous internet connectivity**
- Allows devices to **maintain the same IP address**
- Supports **mobile users moving between networks**
- Maintains **ongoing sessions without interruption**

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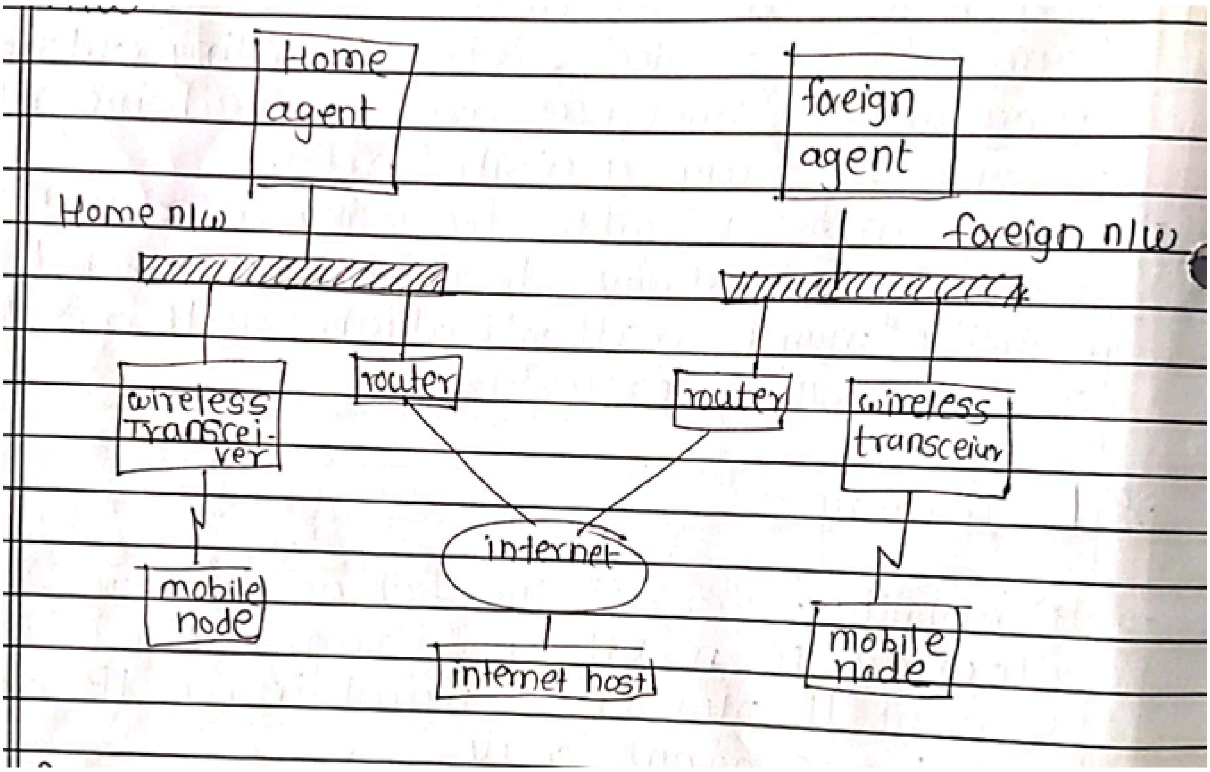
7. Disadvantages

- Additional **delay due to tunneling**
- Increased **network complexity**
- Possible **security vulnerabilities**



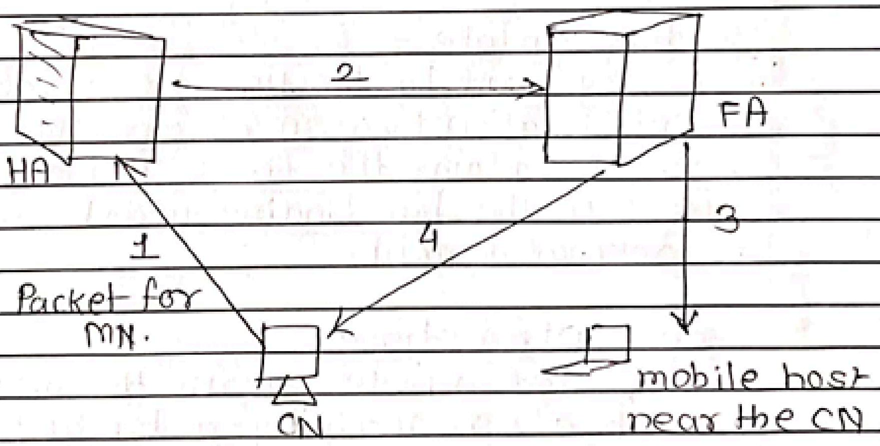
OPTIMIZATION

* Optimization –

→ If the corresponding Node (CN) & MN are very near, then also the IP packet has to travel a long way to reach the MN. This is inefficient behavior of a non optimized mobile IP called is Triangular Routing.

→ The triangle is made of the 3 segments:

i) CN to HA
ii) HA to COA / MN
iii) MN back to CN



→ To solve triangular routing problem, a route optimization protocol has been introduced.

→ Basically this protocol defines some messages as to inform CN of an upto date location of MN.

→ One the current location of the MN is known, the CN itself problems tunneling & sends packet directly to MN.

IPv6 (Internet Protocol Version 6)

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1. Definition

IPv6 (Internet Protocol Version 6) is the **latest version of the Internet Protocol** used to identify devices on a network and allow communication over the Internet.

It was developed to **replace IPv4**, as IPv4 addresses were becoming insufficient due to the rapid growth of internet-connected devices.

Exam Definition:

IPv6 is a **network layer protocol that provides 128-bit addressing to identify devices on the Internet and supports a very large number of unique IP addresses.**

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2. Need for IPv6

IPv4 uses **32-bit addressing**, which provides approximately **4.3 billion unique addresses**.

Due to the rapid growth of the Internet, including **smartphones, computers, servers, and IoT devices**, IPv4 addresses started running out.

IPv6 solves this issue by providing **128-bit addressing**, which allows an extremely large number of unique addresses.

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3. IPv6 Address Format

The length of an IPv6 address is **128 bits**.

It is written as **eight groups of hexadecimal numbers** separated by colons (**:**).

Example of IPv6 Address:

2001:0db8:85a3:0000:0000:8a2e:0370:7334

Rules of IPv6 Addressing

- Uses **hexadecimal numbers (0–9 and A–F)**
- Groups are separated by **colons (:)**
- **Leading zeros can be removed**

Shortened Format Example:

2001:db8:85a3::8a2e:370:7334

The symbol **::** represents one or more groups of zeros.

4. Features of IPv6

Feature	Explanation
Large Address Space	Uses 128-bit addresses, providing a huge number of IP addresses
Improved Security	Built-in support for IPsec for secure communication
Auto Configuration	Devices can automatically configure their IP addresses
Efficient Routing	Simplified header improves routing efficiency
No NAT Required	Every device can have its own unique public IP address

5. IPv4 vs IPv6

Feature	IPv4	IPv6
Address Size	32 bit	128 bit
Address Format	Decimal numbers	Hexadecimal numbers
Example	192.168.1.1	2001:db8::1
Address Capacity	~4.3 Billion	Extremely Large
Security	Optional	Built-in IPsec

6. Advantages of IPv6

- Provides a **very large number of IP addresses**
- Offers **improved security features**
- Enables **faster and more efficient routing**
- Supports **modern internet devices and IoT**
- Eliminates the need for **Network Address Translation (NAT)**

Macro Mobility: MIPv6 (Mobile IPv6)

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1. Definition

Macro Mobility refers to the movement of a **mobile device between different large networks or domains**, such as moving from one city network to another or between different Internet service providers.

MIPv6 (Mobile IPv6) is a protocol designed to support mobility in **IPv6 networks**, allowing a mobile device to **change its location while maintaining the same permanent IP address**.

Exam Definition:

Macro Mobility using **Mobile IPv6 (MIPv6)** allows a mobile node to move between different networks while maintaining continuous communication using its permanent home address.

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2. What is Macro Mobility

Macro mobility occurs when a mobile device moves **between different administrative domains or networks**.

Examples

- Moving from **College Wi-Fi to Mobile Data**
- Traveling from **Mumbai Network to Pune Network**
- Switching between **different Internet service providers**

This type of mobility requires **global mobility support**, which is provided by **Mobile IPv6 (MIPv6)**.

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3. Components of MIPv6

Component	Description
Mobile Node (MN)	A mobile device that moves between different networks
Home Agent (HA)	A router in the home network that keeps track of the mobile node
Correspondent Node (CN)	The device communicating with the mobile node
Care-of Address (CoA)	A temporary IPv6 address assigned to the mobile node in the foreign network

4. Working of MIPv6

Step 1 – Movement of Mobile Node

The **mobile node moves from its Home Network to a Foreign Network**.

Step 2 – Care-of Address Generation

The mobile node obtains a **Care-of Address (CoA)** in the new network.

Step 3 – Binding Update

The mobile node sends a **Binding Update message** to the **Home Agent** and **Correspondent Node** informing them about its new location.

Step 4 – Direct Communication

Packets are sent **directly to the Care-of Address**, which eliminates the problem of **triangular routing**.

5. Features of MIPv6

Feature	Explanation
No Foreign Agent Required	The mobile node manages mobility itself
Route Optimization	Enables direct communication between nodes
Auto Configuration	IPv6 supports automatic address configuration
Improved Security	Built-in IPsec support for secure communication

6. Advantages of MIPv6

- Provides **continuous communication while moving**
- Enables **faster routing using route optimization**
- Eliminates **triangular routing problem**
- Provides **better support for mobile devices**

FMIPV 6

FMIPV6 (Fast Hand over for mobile IPV6) :-

→ Mobile IPV6 (MIPV6) enables a mobile node (MN) to maintain its connectivity to the internet when moving from one access point to another. This process is referred to as 'handover'.

→ During handover, there is a period during which the mobile node is unable to send or receive packets. This period is called 'Hand over latency'.

→ Hand over latency results from the standards' handover procedure such as movement detection new care of addr configuration, binding updates.

→ This Hand over latency is often unacceptable to real-time traffic such as Voice over IP (VoIP)

→ The fast handover for mobile IPV6 (FMIPV6) aims at reducing the long handover latency in mobile IPV6 by fast movement detection & fast binding update.

→ It uses anticipation based on layer 2 trigger information of the mobile node (MN) to obtain a new care-of-addr at the new link while still connected to the previous link, thus reducing handover delay.

→ Furthermore, it also reduces packet loss by buffering before the real link layer handover take place.

Below is a **clean MS Word–ready formatted notes structure** for **Micro Mobility** in **Mobile Computing**. You can **copy–paste it directly into Word** for neat **exam notes or assignment submission**.

Suggested Word Formatting

- **Title:** 20 pt, Bold, Center
- **Main Headings:** 16 pt, Bold
- **Sub Headings:** 14 pt, Bold
- **Body Text:** 12 pt (Calibri / Times New Roman)
- **Line Spacing:** 1.5
- **Alignment:** Justified

Micro Mobility (Mobile Computing)

1. Definition

Micro Mobility refers to the movement of a **mobile device within the same network or administrative domain**, such as moving between different access points inside a campus, office, or local area network.

It focuses on **local mobility management**, ensuring that communication continues smoothly when a device moves within a small geographical area.

Exam Definition:

Micro Mobility is a technique that manages the **movement of a mobile node within the same network domain while maintaining continuous communication.**

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2. What is Micro Mobility

Micro mobility occurs when a mobile device moves **within a single administrative network**.

The mobility management is handled **locally within the network**, without involving the global Internet infrastructure.

Examples

- Moving between **WiFi access points inside a college campus**
- Moving between **different routers in a company network**
- Switching between **cells within the same service provider network**

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3. Purpose of Micro Mobility

Micro mobility is designed to:

- **Reduce handover delay**
- **Reduce signaling traffic in the global network**
- **Improve network efficiency**
- **Provide faster local mobility management**

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4. Working of Micro Mobility

1. A **mobile node moves within the same network domain**.
2. The **local routers manage the location of the device**.
3. The device updates its location **only within the local network**.
4. The **home network does not need to be updated every time** the device moves.

This reduces communication overhead and improves network performance.

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5. Advantages of Micro Mobility

- Faster handover within local networks
- Reduced signaling traffic in global networks
- Lower latency
- Improved network performance

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7. Difference Between Macro Mobility and Micro Mobility

Feature	Macro Mobility	Micro Mobility
Movement Area	Between different networks	Within same network
Scope	Global mobility	Local mobility
Protocol Example	MIPv6	HMIPv6, FMIPv6
Network Update	Global network update required	Only local network update

Below is a **clean MS Word–ready formatted notes structure** for **Cellular IP, HAWAII, and HMIPv6** in **Mobile Computing**. You can **copy–paste directly into Word** for neat **exam notes or assignment submission**.

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Micro Mobility Protocols

(Cellular IP, HAWAII, HMIPv6)

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1. Cellular IP

Definition

Cellular IP is a **micro mobility protocol** designed to support fast handoff and mobility management within a **cellular wireless network**.

It allows mobile devices to move between base stations while maintaining continuous communication with minimal delay.

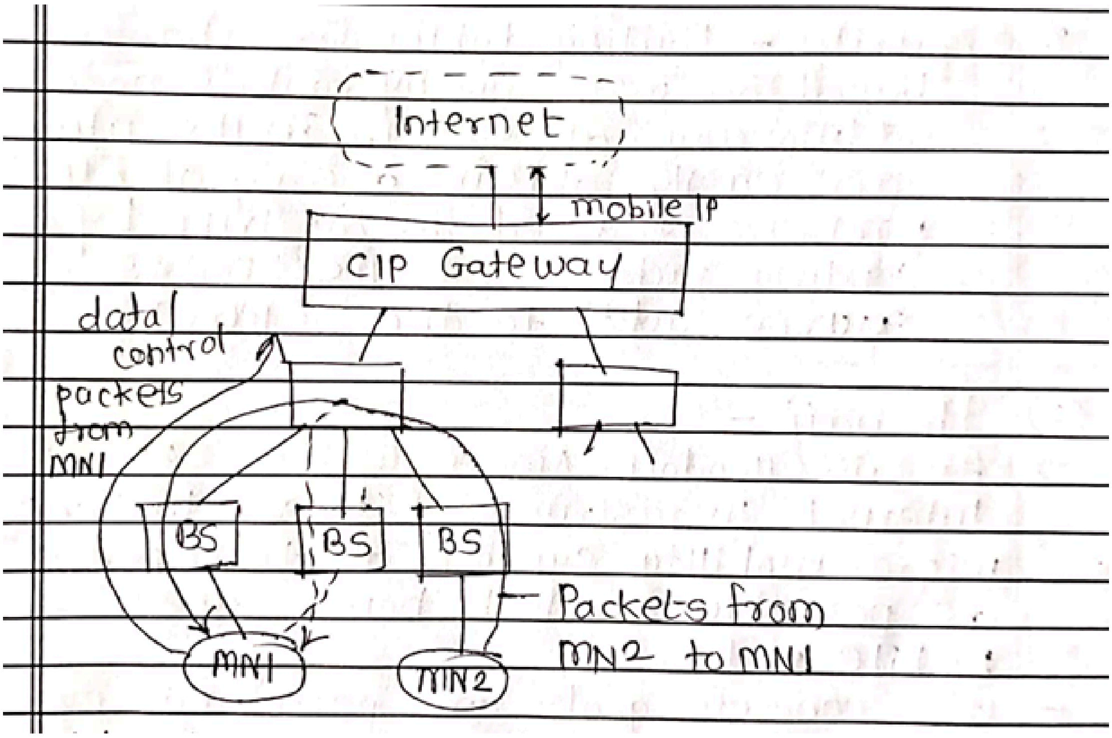
Exam Definition:

Cellular IP is a micro mobility protocol that provides **fast handoff and location management for mobile nodes moving within a cellular network.**

Working of Cellular IP

1. A **mobile node connects to a base station**.
2. The network keeps track of the device using **routing caches**.
3. When the mobile node moves to another base station:
   - The routing cache is **updated automatically**.
4. Data packets are forwarded through the **updated routing path**.

This allows **fast packet delivery without global updates**.



Advantages

- Fast handover
- Reduced network delay
- Efficient routing
- Suitable for cellular wireless networks

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2. HAWAII (Handoff-Aware Wireless Access Internet Infrastructure)

Definition

HAWAII is a **micro mobility protocol** that manages the movement of mobile nodes within a **single administrative domain**.

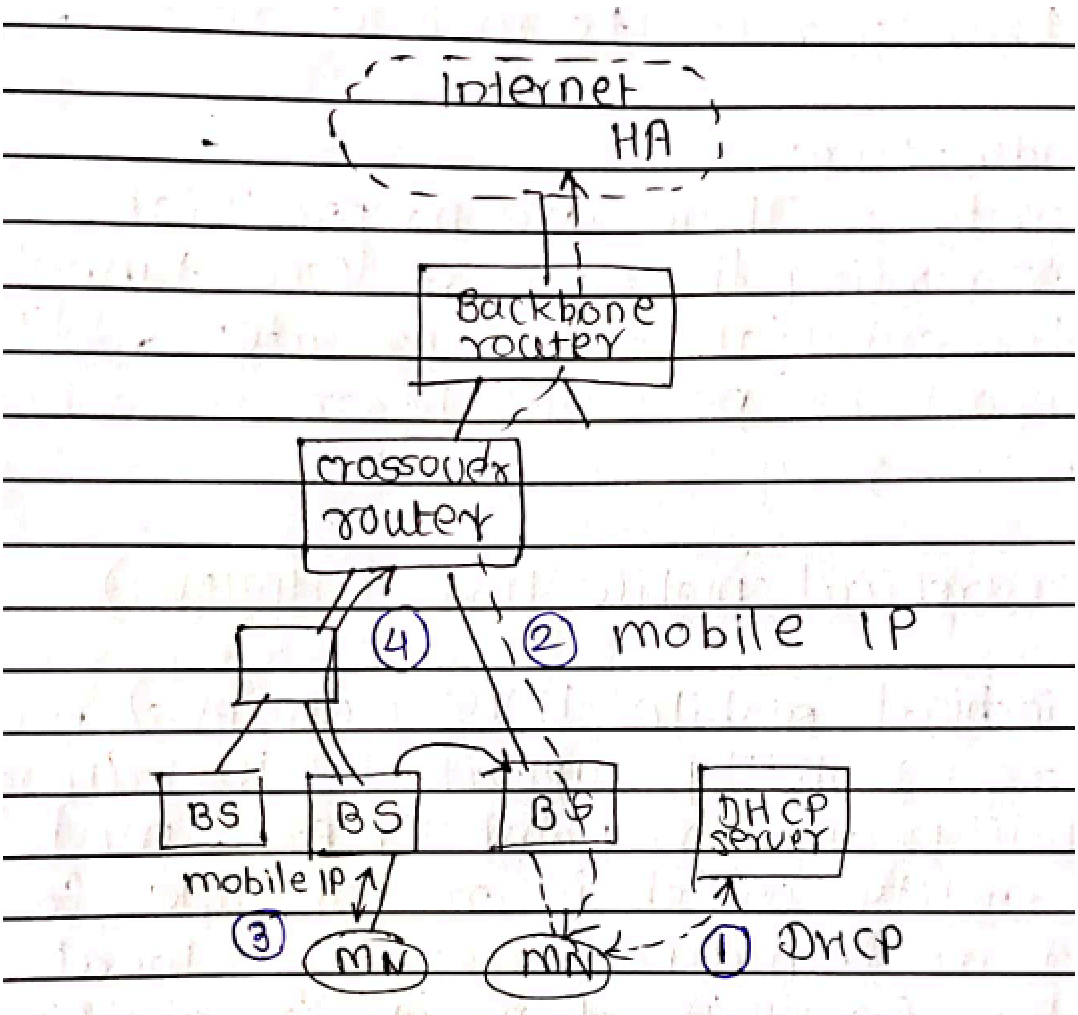
It supports fast handoffs by updating routing paths inside the network.

Exam Definition:

HAWAII is a micro mobility protocol that supports **fast handover and efficient routing for mobile nodes within a local network domain.**

Working of HAWAII

1. Mobile node connects to the **access router**.
2. The network creates a **routing path** from the gateway to the mobile node.
3. When the node moves to another access router:
    - Only **local routing tables are updated**.
4. Data packets follow the **updated path to the new location**.



Advantages

- Faster mobility management
- Less signaling overhead

* Efficient packet delivery
* Supports seamless connectivity

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3. HMIPv6 (Hierarchical Mobile IPv6)

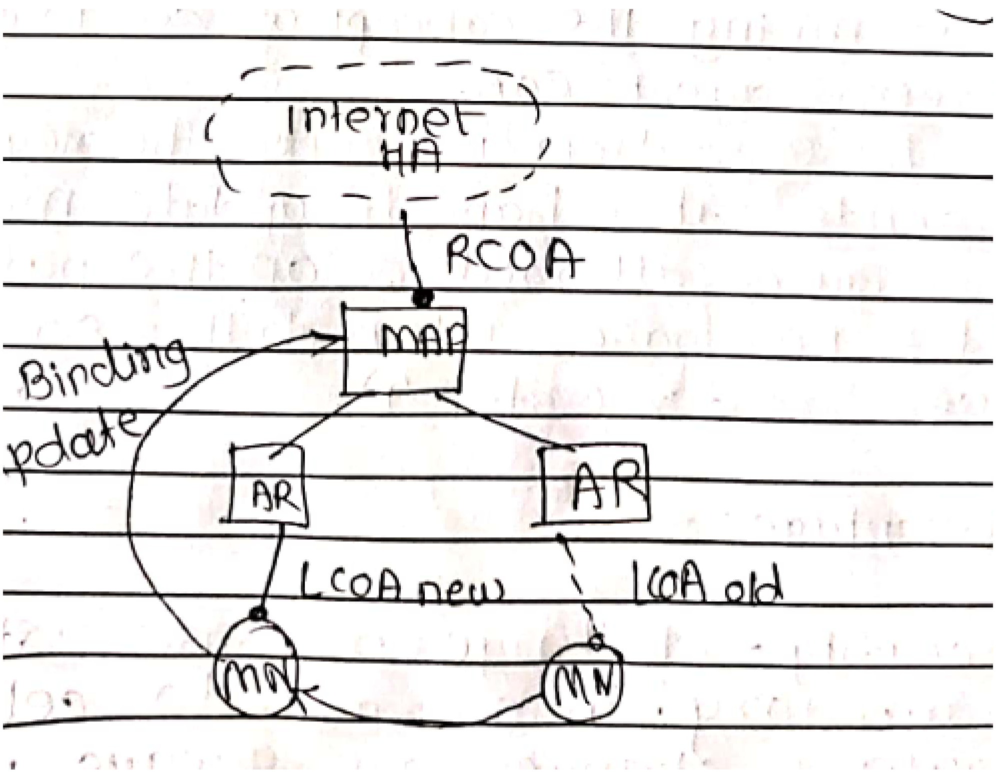
Definition

HMIPv6 (Hierarchical Mobile IPv6) is an extension of **Mobile IPv6** designed to improve **micro mobility management**.

It introduces a hierarchical structure that reduces the number of updates sent to the home agent.

Exam Definition:

HMIPv6 is a mobility management protocol that uses a **hierarchical architecture to manage local mobility and reduce signaling overhead in Mobile IPv6 networks.**



Working of HMIPv6

1. Mobile node enters a **MAP domain**.

2. The MAP assigns a **Regional Care-of Address (RCoA)**.
3. The mobile node moves within the same MAP domain.
4. Only the **MAP updates the location**, not the home agent.

This reduces signaling traffic and improves mobility performance.

Features

Feature	Explanation
Hierarchical Structure	Uses Mobility Anchor Point
Reduced Signaling	Fewer updates to Home Agent
Local Mobility Support	Efficient micro mobility
Improved Scalability	Better network performance

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4. Comparison of Micro Mobility Protocols

Feature	Cellular IP	HAWAII	HMIPv6
Mobility Type	Micro Mobility	Micro Mobility	Micro Mobility
Mobility Management	Routing cache	Domain-based routing	Hierarchical
Main Component	Base stations	Access routers	Mobility Anchor Point
Network Update	Local update	Local routing update	MAP-based update

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